[January] REGULAR EXAMINATIONS
[B.Tech/B.E]
[Information Technology, Computer Science and Engineering, Artificial Intelligence and Data Science, Computer Science and Design and Artificial Intelligence and Machine Learning]
[IV] Semester
Regulations [R23 V1.1]

COURSE CODE	COURSE TITLE
2311CSS403J	Data Science Essentials

CO1: Explain the fundamentals, scope, and lifecycle of Data Science and its applications

CO2: Apply appropriate techniques to prepare datasets for analysis.

CO3: Analyse data using statistical measures and probability concepts to extract meaningful insights

CO4: Implement machine learning techniques for data analytics tasks.

CO5: Interpret data using visualization techniques, discuss ethical issues and practical applications of Data Science.

K1 – Remember K2 – Understand K3 – Apply K4 – Analyse K5 – Evaluate K6 – Create

Unit – I

Part A

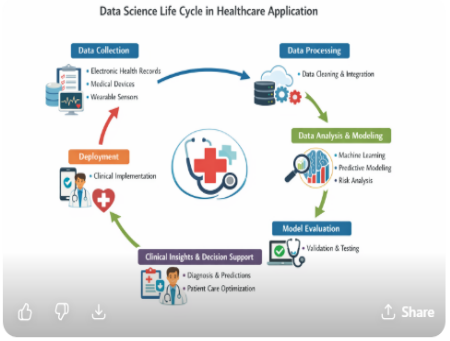
Q.No	Questions	CO's	Unit	Bloom's Level
1.	Define Data Science. Why is Data Science required?	CO1	1A	K1
1 Ans	Data science is the study of data used to extract meaningful insights for business decisions. It's required because it transforms raw data into actionable intelligence, helping organizations optimize processes, understand customers, forecast performance, personalize experiences, and gain a competitive edge in today's data-rich world.	CO1	1A	K1
2.	State the scope of Data Science in airline systems.	CO1	1A	K1
2 Ans	The scope of Data Science in airlines includes optimizing operations and improving customer experience. Key applications are predicting flight delays, dynamic ticket pricing, predictive maintenance of aircraft, and personalized customer offers.	CO1	1A	K1
3.	List the key components of Data Science.	CO1	1A	K1
3 Ans	The key components of data science: Data	CO1	1A	K1

<table>
<tr><td></td><td>Collection, Data Cleaning, Data Exploration & Visualization, Data Modeling, and Model Evaluation & Deployment</td><td></td><td></td><td></td></tr>
<tr><td>4.</td><td>Identify the major stages in the evolution of Data Science.</td><td>CO1</td><td>1A</td><td>K2</td></tr>
<tr><td>4 Ans</td><td>The major stages in the evolution of Data Science are:
1. Early Data Handling (Traditional Analysis)
2. Business Intelligence (BI Phase)
3. Big Data Emergence (Hadoop/NoSQL)
4. Rise of Data Science (Predictive Analytics)
5. AI-Centric Data Science (Deep Learning & Automation)</td><td>CO1</td><td>1A</td><td>K2</td></tr>
<tr><td>5.</td><td>Define structured data with one example.</td><td>CO1</td><td>1A</td><td>K1</td></tr>
<tr><td>5 Ans</td><td>Structured data is information organized in a predefined format or model, such as rows and columns in a database or spreadsheet, making it easily searchable and analyzable by both humans and machines.
Example: Customer table</td><td>CO1</td><td>1A</td><td>K1</td></tr>
<tr><td>6.</td><td>Outline semi-structured data with example.</td><td>CO1</td><td>1A</td><td>K2</td></tr>
<tr><td>6 Ans</td><td>Semi-structured data is data that does not reside in a traditional relational database (like SQL) but still has some organizational properties, such as tags or markers, that make it easier to analyze than completely unstructured data.
Examples:
JSON (JavaScript Object Notation)
XML (eXtensible Markup Language)
CSV files with inconsistent rows
Emails (with structured headers and unstructured body text)</td><td>CO1</td><td>1A</td><td>K2</td></tr>
<tr><td>7.</td><td>Describe unstructured data with one example.</td><td>CO1</td><td>1A</td><td>K1</td></tr>
<tr><td>7 Ans</td><td>Unstructured data refers to information that does not have a predefined format or structure. It is messy, unorganized and hard to sort.
Examples: text documents, images, audio files, videos and social media posts</td><td>CO1</td><td>1A</td><td>K1</td></tr>
<tr><td>8.</td><td>Recall the main objective of the Data Science life cycle.</td><td>CO1</td><td>1A</td><td>K1</td></tr>
<tr><td>8 Ans</td><td>The main objective of the Data Science life cycle is to transform raw data into actionable insights and predictive models to solve business problems and drive informed decision-making.</td><td>CO1</td><td>1A</td><td>K1</td></tr>
<tr><td>9.</td><td>How Data Science differs from Statistics?</td><td>CO1</td><td>1A</td><td>K2</td></tr>
<tr><td>9 Ans</td><td><table><tr><th>Feature</th><th>Data Science</th><th>Traditional Statistics</th></tr><tr><td>Focus</td><td>Building models, extracting insights, making</td><td>Hypothesis testing, drawing conclusions, understanding</td></tr></table></td><td></td><td></td><td></td></tr>
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<table>
<tr><td></td><td><table>
<tr><td></td><td>predictions</td><td>relationships</td></tr>
<tr><td>Data size</td><td>Often deals with large, complex datasets (big data)</td><td>Can work with smaller, focused datasets</td></tr>
<tr><td>Typical output</td><td>Predictive model, actionable insights, data visualizations</td><td>Statistical tests, p-values, confidence intervals</td></tr>
<tr><td>Tools & Techniques</td><td>Programming languages (Python, R), machine learning algorithms, data visualization tools</td><td>Statistical software (R, SAS), probability theory, hypothesis testing frameworks</td></tr>
</table></td><td>CO1</td><td>1A</td><td>K2</td></tr>
<tr><td>10.</td><td>Differentiate between Data Science and Machine Learning.</td><td>CO1</td><td>1A</td><td>K2</td></tr>
<tr><td>10 Ans</td><td><table>
<tr><th>Feature</th><th>Data Science</th><th>Machine Learning</th></tr>
<tr><td>Scope & Application</td><td>Broad covers data collection, cleaning, analysis, visualization and modelling</td><td>Narrower focuses only on building predictive models</td></tr>
<tr><td>Techniques</td><td>Statistics, data analysis, visualization, ML, business intelligence</td><td>Algorithms like regression, decision trees, clustering, neural networks</td></tr>
<tr><td>Goal</td><td>Extract insights and support decision-making</td><td>Automate predictions and pattern recognition</td></tr>
<tr><td>Output</td><td>Reports, dashboards, insights, models</td><td>Predictive or classification models</td></tr>
</table></td><td>CO1</td><td>1A</td><td>K2</td></tr>
<tr><td>11.</td><td>Explain how Artificial Intelligence is related to Data Science.</td><td>CO1</td><td>1A</td><td>K2</td></tr>
<tr><td>11 Ans</td><td><table>
<tr><th>Artificial Intelligence (AI)</th><th>Data Science</th></tr>
</table></td><td>CO1</td><td>1A</td><td>K2</td></tr>
</table>

	<table><tr><td>Focuses on creating systems that mimic human intelligence.</td><td>Analyses data to extract insights for decision-making.</td></tr><tr><td>Build autonomous systems that learn and act.</td><td>Uncover patterns and trends from data.</td></tr><tr><td>Machine learning, deep learning, NLP, neural networks.</td><td>Statistics, data visualization, Python, R, SQL.</td></tr><tr><td>Self-driving cars, chatbots, robotics, image recognition.</td><td>Business analytics, predictive modelling, dashboards.</td></tr><tr><td>Automation and intelligent decision-making.</td><td>Data analysis and interpretation.</td></tr><tr><td>Uses data to train models for predictions or actions.</td><td>Uses data to generate insights and reports.</td></tr></table>			
12.	Summarize the role of data collection in the Data Science life cycle	CO1	1A	K2
12 Ans	Data collection is the foundational step in the data science lifecycle. It involves gathering raw data from various sources to address specific business questions or hypotheses. The quality and relevance of the data collected significantly impact the subsequent stages of the data science process.	CO1	1A	K2
13.	Describe any four real-world applications of Data Science.	CO1	1A	K2
13 Ans	1. Healthcare: Data science identifies and predicts disease, and personalizes healthcare recommendations. 2. Transportation: Data science optimizes shipping routes in real-time. 3. Sports: Data science accurately evaluates athletes' performance. 4. E-commerce: Data science automates digital ad placement.	CO1	1A	K2
14.	Give the importance of data pre-processing in Data Science.	CO1	1A	K2
14 Ans	Data preprocessing is the process of cleaning, transforming, and organizing raw data into a structured format for analysis, machine learning (ML), and artificial intelligence (AI) applications. It helps to make data more accurate, consistent, and suitable for modeling by removing noise, handling missing values, and standardizing features.	CO1	1A	K2
15.	Outline any two challenges faced in Data Science	CO1	1A	K2
15 Ans	1. Data Availability and Quality 2. Data Integration	CO1	1A	K2

Part- B

Q.No	Questions	CO's	Unit	Bloom's Level
1.	Apply the Data Science life cycle to a healthcare application.	CO1	1B	K3
1 Ans	1. Problem Definition & Business Understanding **(2 marks)** 2. Data Acquisition & Collection **(2 marks)** 3. Data Cleaning & Preparation **(2 marks)** 4. Exploratory Data Analysis (EDA) **(2 marks)** 5. Modeling & Algorithm Development. **(2 marks** 6. Evaluation & Validation **(1 marks)** 7. Deployment & Monitoring **(2 marks)** 8. Diagram **(2)** 	CO1	1B	K3
2.	Categorize the evolution of Data Science with suitable examples.	CO1	1B	K4
2 Ans	Evolution of Data Science 1. Early Data Handling and Traditional Analysis **(2 marks)** 2. Business Intelligence (BI) Phase **(2 marks)** 3. Big Data Emergence **(2 marks)** 4. Rise of Data Science **(2 marks)** **5.** Machine Learning–Driven Evolution **(2 marks)** **6.** AI-Centric Data Science (Current Trend) **(2 marks)** 7. Example **(1 marks)**	CO1	1B	K4
3.	Apply Data Science techniques in business decision-making scenarios.	CO1	1B	K3
3 Ans	1. **Descriptive Analytics (What happened?): (3 marks)** Technique: Aggregation and Visualization. 2. **Diagnostic Analytics (Why did it happen?): (3 marks)** Technique: Drill-down analysis and correlations. 3. **Predictive Analytics (What will happen?): (3 marks)** Technique: Regression models and forecasting. be proactive rather than reactive (e.g., stocking up early). 4. **Prescriptive Analytics (How can we make it**	CO1	1B	K3

	happen?): (4 marks) Technique: Optimization algorithms. Scenario: An algorithm automatically adjusts prices and orders inventory to ensure the July spike is profitable.			
4.	Machine Learning is a subset of AI, and Data Science is the tool to apply them." Critically analyze this statement with a detailed comparison of AI, ML, and Data Science.	CO1	1B	K4
4 Ans	1. **AI as the Broad "Goal" (The What):(4 marks)** Example: A chess computer that calculates moves. 2. M**achine Learning as the "Subset" (The How): (4 marks)** ML is a specific subset of AI where machines are not explicitly programmed with rules. Instead, they "learn" patterns from data. 3. **Data Science as the "Tool" (The Process): (5 marks)** Data Science is the practical field that applies these theories to the real world. A pure ML algorithm is useless without clean data.	CO1	1B	K4
5.	Analyse the characteristics and use cases of structured, semi-structured, and unstructured data.	CO1	1B	K4
5 Ans	1. **Structured Data (4 marks)** Examples & Use Cases: ● Customer Databases: Tables storing clear, distinct values like First Name, Last Name, Phone Number, and Email Address. ● Financial Records: Bank transaction logs where every entry has a timestamp, amount, and account ID. 2. **Semi-Structured Data(4 marks)** Examples & Use Cases: ● Web Data Exchange: JSON (JavaScript Object Notation) and XML (eXtensible Markup Language) are the primary formats for sharing data between servers and web applications. ● Emails: These are considered semi-structured because they have structured headers (To, From, Date) but an unstructured body text. ● Data Logs: CSV files or server logs that may contain inconsistent rows. 3. **Unstructured Data (5 marks)** Examples & Use Cases: ● Multimedia: Images, audio files, and video content. ● Social Media: Posts, comments, and tweets which contain slang, hashtags, and varying languages. ● Documents: Word documents, PDFs, and slide presentations.	CO1	1B	K4
6.	**Examine the role of Data Science in autonomous cars.**	CO1	1B	K4

6 Ans	**1.Characteristics** (4 marks) **a)** Perception and Environment Understanding Techniques used: b) Localization and Mapping c) Decision-Making and Path Planning **2. Data Sources Used in Autonomous Cars (4 marks)** ● Sensor data (camera, LiDAR, radar, ultrasonic sensors) ● Traffic and weather data ● Historical driving data ● Real-time road and infrastructure data **3. Benefits of Data Science in Autonomous Driving (3 marks)** ● Improved road safety and reduction in human error ● Efficient traffic management and reduced congestion **4. Challenges and Limitations (2 marks)** 1. Requirement for massive volumes of high-quality data 2. Difficulty in handling rare and unpredictable scenarios	CO1	1B	K4
7.	Evaluate the major challenges and limitations of Data Science in real-world applications.	CO1	1B	K5
7 Ans	**1. Data-Related Challenges (3 marks)** a) Data Quality Issues b) Data Availability and Integration **2. Technical and Model-Related Limitations) (2 marks)** Scalability and Performance b) Model Interpretability c) Generalization and Overfitting **3. Ethical, Legal, and Social Challenges (2 marks)** a) Bias and Fairness b) Privacy and Data Security c) Ethical Decision-Making **4. Organizational and Human Challenges (2 marks)** a) Skill Gap b) Cost and Infrastructure c) Cultural Resistance **5. Deployment and Maintenance Issues (2 marks)** a) Model Deployment b) Model Monitoring and Maintenance **6. Domain-Specific Limitations (2 marks)** ● Lack of domain understanding can lead to incorrect assumptions ● Models may fail in rare or extreme scenarios ● Results may not be transferable across industries or regions	CO1	1B	K5
8.	Compare and examine the stages of the Data Science life cycle with suitable examples.	CO1	1B	K4

8 Ans	**1. Problem Definition: (2 marks)** ● Action: Identifying the business. ● Example: A hospital wants to reduce patient readmission rates. **2. Data Collection (Acquisition): (2 marks)** ● Action: Gathering raw data from diverse sources like sensors, logs, or APIs. ● Example: Collecting Electronic Health Records (EHR) and real-time vitals from IoT devices. **3. Data Cleaning (Preparation): (2 marks)** ● Importance: This often consumes 80% of the project time. **4. Exploratory Data Analysis (EDA): (2 marks)** ● Example: Plotting patient age vs. disease severity to see if older patients are at higher risk. **5. Modeling: (2 marks)** ● Example: Training a model to predict "Yes/No" for sepsis based on blood test results. **6. Evaluation: (2 marks)** ● Example: Checking if the sepsis model has too many "False Negatives" (missing sick patients). **7. Deployment & Monitoring: (1 marks)** ● Example: Creating an alert system that notifies doctors instantly on their tablets.	CO1	1B	K4

Part- C

Q.No	Questions	CO's	Unit	Bloom's Level
1.	Evaluate the Scope and applications of Data Science in finance and education, considering its challenges and limitations.	CO1	1C	K5
1 Ans	**Scope of Data Science(3)** Data Science involves extracting insights from structured and unstructured data using statistics, machine learning, and computational techniques. Its scope includes: **Applications of Data Science in Finance(4 marks)** a) Risk Management b) Investment and Trading c) Customer Analytics d) Operations and Compliance **Applications of Data Science in Education (4 marks)** a) Learning Analytics b) Personalized Learning c) Institutional Planning d) Assessment and Feedback **Challenges and Limitations(4 marks)** a) Data Quality and Availability b) Ethical and Privacy Concerns c) Bias and Fairness d) Interpretability and Trust e) Skill and Infrastructure Constraints	CO1	1C	K5

2.	Imagine you are hired by an e-commerce company to reduce customer churn. Describe the Data Science Life Cycle steps you would take to solve this problem.	CO1	1C	K5
2 Ans	**Step 1: Problem Definition & Business Understanding (2 marks)** ● Goal**:** Identify customers likely to stop buying (churn) within the next 30 days. **Step 2: Data Collection (2 marks)** ● Behavioral Data: Website visits, time spent on pages, and cart abandonment (from Web Logs). **Step 3: Data Cleaning & Preparation (2 marks)** ● Feature Engineering: Create specific features that signal churn, such as: **Step 4: Exploratory Data Analysis (EDA) (2 marks)** *Hypothesis:* "Users who complain more than twice in a month are 80% likely to churn." **Step 5: Modeling (3 marks)** ● Algorithm: I would use Logistic Regression (for simple probability scores) or Random Forest (for high accuracy). ● Training: Train the model on historical data (e.g., last year's user behavior). **Step 6: Evaluation (2 marks)** ● *Reasoning:* It is better to falsely flag a happy customer (and give them a discount) than to miss an unhappy customer who is about to leave. We want to minimize "False Negatives." **Step 7: Deployment & Action (2 marks)** ● **Integration:** The model runs nightly on the active user base.	CO1	1C	K5

Unit- 2

Part A

Q.No	Questions	CO's	Unit	Bloom's Level
1.	State the difference between primary and secondary data sources?	CO2	2A	K2
1 Ans	Difference between Primary and Secondary data sources ● Primary data: Collected first-hand for a specific purpose (e.g., surveys, experiments). ● Secondary data: Already collected by others for a different purpose (e.g., reports, databases).	CO2	2A	K2
2.	Identify two data quality issues and suggest a technique to resolve them.	CO2	2A	K2
2 Ans	1. Inconsistency: Resolved by Data Standardization.	CO2	2A	K2

	2. Missing Values: Resolved by Imputation (replacing missing values with mean/median/mode).			
3.	Define a Linear Congruential Generator (LCG).	CO2	2A	K1
3 Ans	Linear Congruential Generator (LCG) An LCG is a method for generating pseudo-random numbers using the recurrence relation	CO2	2A	K1
4.	A dataset contains a 'Temperature' column where some values are recorded in Celsius and others in Fahrenheit. Which data quality issue does this primarily represent explain the issues?	CO2	2A	K2
4 Ans	Data inconsistency-In data quality, Data Consistency refers to the requirement that data values across different datasets, databases, or even within the same record do not contradict each other. It ensures that "one version of the truth" exists across an entire organization. Consistency issues usually arise when data is moved, replicated, or updated in one place but not another.	CO2	2A	K2
5.	How does the 'Binning' technique reduce noise?.	CO2	2A	K2
5 Ans	Binning smooths data by distributing it into small "bins." The values in each bin are then replaced by the mean or median of that bin, reducing the impact of random fluctuations.	CO2	2A	K2
6.	Why is "Handling Missing Data" critical in the data science life cycle?	CO2	2A	K1
6 Ans	Many machine learning algorithms cannot handle missing values; ignoring them can lead to biased results, loss of information, and inaccurate model performance**.**	CO2	2A	K1
7.	Explain the difference between Data Normalization and Standardization.	CO2	2A	K2
7 Ans	Normalization: Scales data to a fixed range (e.g., 0 to 1). Standardization: Transforms data to have mean 0 and standard deviation 1.	CO2	2A	K2
8.	Why is Data Cleaning considered a critical step in the KDD process?	CO2	2A	K2
8 Ans	Data cleaning improves data quality, leading to more accurate and reliable knowledge discovery results.	CO2	2A	K2
9.	Differentiate between Feature Selection and Feature Extraction.	CO2	2A	K2
9 Ans	Feature Selection: Selects existing relevant features. Feature Extraction: Creates new features by transforming original ones.	CO2	2A	K2
10.	How a "Box Plot" helps in outlier detection.	CO2	2A	K2
10 Ans	A Box Plot detects outliers by using the Interquartile Range (IQR) to establish statistical boundaries called "fences." It identifies any data point that falls significantly outside the central distribution of the data.	CO2	2A	K2

11.	Given the sorted data: 4, 8, 15, 21, 21, 24, 25, 28, 34, smooth the data using bin means with a bin depth of 3.	CO2	2A	K3
11 Ans	Resulting Smoothed Data: 9, 9, 9, 22, 22, 22, 29, 29, 29	CO2	2A	K3
12.	What are the implications of choosing a small modulus (m) in an LCG?	CO2	2A	K1
12 Ans	A small modulus leads to a shorter period and poor randomness.	CO2	2A	K1
13.	Why is data transformation necessary before applying certain machine learning algorithms?	CO2	2A	K2
13 Ans	It ensures data is on a common scale, improves performance, and meets algorithm assumptions.	CO2	2A	K2
14.	Given a dataset {10, 20, 30, 40, 50}, calculate the Min-Max Normalized value for 20 in the range [0, 1].	CO2	2A	K3
14 Ans	The Min-Max normalized value for 20 is 0.25.	CO2	2A	K3
15.	Using $X_{n+1} = (3X_{n+2}) \bmod 7$ with seed $X_0 = 1$, find the next two numbers (X1, X2).	CO2	2A	K3
15 Ans	Ans: $X_1 = 5$, $X_2=3$	CO2	2A	K3

Part- B

Q.No	Questions	CO's	Unit	Bloom's Level
1.	Categorize the various data sources and data collection methods. Illustrate how appropriate data collection methods can be applied to ensure data quality in a real-world data science application.	CO2	2B	K4
1 Ans	**Classification of Data Sources** **1. Primary Data Sources (3 marks)** Primary data is collected **directly from the original source** for a specific purpose. **Examples:** ● Surveys and questionnaires ● Interviews and focus groups ● Direct observations ● Controlled experiments ● Sensor and IoT-generated data **2. Secondary Data Sources (3 marks)** Secondary data is data that has already been collected for another purpose. **Examples:** ● Government census data ● Company sales records ● Research papers and journals ● Public datasets (Kaggle, WHO, UCI) ● Inconsistent formats **3. Data Collection Methods (3 marks)** **1.** Surveys and Questionnaires 2. Interviews	CO2	2B	K4

<table>
<tr><td></td><td>3. Observations
4. Automated Data Collection
● High accuracy
● Real-time
● Technical complexity
Ensuring Data Quality – Real World Application(4 marks)
Example: E-commerce Recommendation System
<table><tr><td>Data Quality Dimension</td><td>Data Collection Strategy</td></tr><tr><td>Accuracy</td><td>Automated transaction logging</td></tr><tr><td>Completeness</td><td>Mandatory user fields</td></tr><tr><td>Consistency</td><td>Standardized formats</td></tr><tr><td>Timeliness</td><td>Real-time clickstream data</td></tr><tr><td>Reliability</td><td>Validation checks</td></tr></table></td><td></td><td></td><td></td></tr>
<tr><td>2.</td><td>Define data cleaning. Analyze the common data cleaning techniques used to improve dataset quality, illustrating how each technique addresses specific data quality issues with suitable examples.</td><td>CO2</td><td>2B</td><td>K4</td></tr>
<tr><td>2 Ans</td><td>Handling missing data(4 marks)
One major data cleaning technique is handling missing values. Missing values may occur due to incomplete data entry or system failures. Common techniques include mean, median, or mode imputation. For example, missing salary values can be replaced with the average salary of employees.
Duplicate data removal (4 marks)
Another technique is duplicate data removal. Duplicate records occur when the same data entry is recorded multiple times, leading to biased analysis. For instance, duplicate customer records can inflate sales figures. Identifying and removing duplicates ensures data uniqueness.
Noise reduction (3 marks)
It is also an important cleaning step. Noise refers to random errors or fluctuations in data. Techniques such as binning, moving average, and regression smoothing help reduce noise. For example, sensor data fluctuations can be smoothed using moving averages.
Inconsistency correction (2 Marks) involves standardizing data formats and resolving conflicting values. For instance, date formats like DD/MM/YYYY and MM/DD/YYYY are standardized to maintain consistency.</td><td>CO2</td><td>2B</td><td>K4</td></tr>
<tr><td>3.</td><td>What is missing data? Classify the types of missing data and explain different methods for handling missing values in data preprocessing.</td><td>CO2</td><td>2B</td><td>K4</td></tr>
<tr><td>3 Ans</td><td>Classification of Missing Data (3 Marks)
Statisticians categorize missing data based on the relationship between the missing values and the</td><td>CO2</td><td>2B</td><td>K4</td></tr>
</table>

	observed data. Understanding these types is crucial for choosing the right handling method. ● Missing Completely at Random (MCAR): ● Missing at Random (MAR): ● Missing Not at Random (MNAR): **2. Methods for Handling Missing Values (7 Marks)** A. Deletion Methods 1. Listwise Deletion (Complete Case Analysis): 2. Pairwise Deletion: 3. Variable Dropping: B. Imputation Methods (Statistical) 1. Mean/Median/Mode Imputation: 2. Constant Value Imputation: 3. Forward/Backward Fill (Time Series): C. Advanced Imputation (Model-Based) 1. K-Nearest Neighbors (KNN): 2. Multiple Imputation (MICE): **3. The Selection Workflow (3 Marks)** Choosing a method depends on the percentage of missing data and the type of data (Categorical vs. Numerical).			
4.	Determine the working of Linear Congruential Generators (LCG). Explain how random number generation is applied in data sampling and simulation tasks.	CO2	2B	K5
4 Ans	**A.Working of Linear Congruential Generators (7 Marks)** **1. Key Parameters Explained** ● (The Seed) ● (The Modulus) ● (The Multiplier) ● (The Increment): **2. Applications in Data Science** Pseudo-random numbers are fundamental to the "Science" part of Data Science, particularly in the following areas: **B. Data Sampling (3 Marks)** ● **Simple Random Sampling:** ● **Train-Test Split:**. ● Bootstrapping: **C. Simulation Tasks (3 Marks)** ● **Monte Carlo Simulations:** These rely on repeated random sampling to obtain numerical results. For example, simulating the price of a stock 10,000 times to predict future risk. ● **Stochastic Processes:** Used in modeling systems with uncertainty, such as queuing at a bank or the spread of a virus through a population. ● **A/B Testing:** Randomly assigning users to a "Control" group or a "Variant" group to test the impact of a new feature.	CO2	2B	K5

5.	Define noise in data. Analyze various noise detection and smoothing techniques used in data preprocessing, highlighting how each technique identifies and reduces noise in datasets.	CO2	2B	K4
5 Ans	**1**. **Defining Noise in Data (2 Marks)** Noise is generally categorized into two types: ● Attribute Noise: Errors in the values of specific variables (e.g., a person's age listed as -5 or 200). ● Class/Label Noise: Errors in the categorical assignment (e.g., a "Spam" email incorrectly labeled as "Not Spam"). **2. Noise Detection Techniques (5 Marks)** Before we can smooth the data, we have to find the culprits. Here is how we identify noise: A. Statistical Outlier Detection B. Clustering C. Visualization **3. Data Smoothing Techniques (6 Marks)** A. Binning ● How it works: Sort the data and distribute it into "bins" (buckets). ● Smoothing Method: * Smoothing by bin means: Replace all values in a bin with the average value of that bin. B. Regression ● How it works: Fits the data points to a mathematical function, such as a linear equation $y = mx + c$. C. Moving Average (Filter) ● How it works: Commonly used in time-series data. It calculates the average of a sliding window of data points.	CO2	2B	K4
6.	Elucidate data transformation and discuss its need in data pre-processing. Illustrate the application of common data transformation techniques used to prepare data for analysis and machine learning tasks	CO2	2B	K3
6 Ans	**Defining Data Transformation (2 Marks)** Data transformation is the process of changing the structure, format, or values of data. While data cleaning removes "bad" data, transformation reshapes "good" data so that its mathematical properties (like scale and distribution) align with the requirements of specific machine learning models. **2. The Need for Data Transformation (3 Marks)** Why can't we just use raw data? ● Unit Inconsistency: ● Algorithm Requirements: ● Normalization of Distribution: ● Improved Convergence: 3. Common Data **Transformation Techniques (8 Marks)**	CO2	2B	K3

	A. Normalization (Min-Max Scaling) This scales the data into a fixed range, usually [0, 1]. It is vital for algorithms that do not assume a specific distribution. ● **Application:** Used in Neural Networks and Image Processing where pixel values are scaled between 0 and 1. **B. Standardization (Z-score Normalization)** ● **Application:** Essential for Principal Component Analysis (PCA) and Clustering, as it handles outliers better than Min-Max scaling. **C. Discretization (Binning)** This involves transforming continuous numeric data into discrete "bins" or intervals. ● **Application:** Converting "Age" (e.g., 18, 25, 60) into categories like "Teen," "Adult," and "Senior." This reduces the complexity of the model and helps in handling noisy data. **D. Concept Hierarchy Generation** Data is transformed from a low-level concept to a higher-level concept. **E. Attribute Construction (Feature Engineering)** New attributes are constructed from existing ones to help the mining process.			
7.	Define data normalization. Evaluate various normalization techniques and critically compare them with data standardization, highlighting their suitability and applications in different data analysis and machine learning scenarios.	CO2	2B	K5
7 Ans	**1.Defining Data Normalization (2 Marks)** Data normalization is the process of scaling numeric data to a specific, narrow range (commonly $[0, 1]$ or $[-1, 1]$) without distorting differences in the ranges of values. Its primary goal is to eliminate **"**magnitude bias**,"** where features with naturally larger numbers (like Annual Income) dominate features with smaller numbers (like Age) during mathematical computations. **2. Evaluation of Normalization Techniques (5 Marks)** **A. Min-Max Scaling** ● **Mechanism:** Linearly transforms the data. The minimum value of the feature becomes 0, and the maximum becomes 1. **B. Decimal Scaling** ● **Mechanism:** Moves the decimal point of values based on the absolute maximum value in the dataset. ● **Suitability:** Useful for quickly scaling data where the range is extremely large, though it is used less frequently in modern ML than Min-Max.	CO2	2B	K5

<table>
<tr>
<td></td>
<td>
3. Comparison: Normalization vs. Standardization (6 Marks)
While both techniques rescale data, they serve different masters depending on the data's "personality."
<table>
<tr><th>Feature</th><th>Normalization (Min-Max)</th><th>Standardization (Z-Score)</th></tr>
<tr><td>Formula</td><td>(x - Min) / (Max - Min)</td><td>(x - \mu) / \sigma</td></tr>
<tr><td>Output Range</td><td>Strictly bounded (e.g., 0 to 1)</td><td>Unbounded (centered at 0)</td></tr>
<tr><td>Outlier Sensitivity</td><td>High. A single outlier can "squish" all other data points into a tiny range.</td><td>Low. It uses the mean and standard deviation, making it more robust.</td></tr>
<tr><td>Distribution</td><td>Does not assume any distribution.</td><td>Assumes/forces a Gaussian (Normal) distribution.</td></tr>
</table>
</td>
<td></td><td></td><td></td>
</tr>
<tr>
<td>8.</td>
<td>Summarize the concept of feature selection. Analyze different feature selection methods and examine their role in improving model performance and reducing model complexity.</td>
<td>CO2</td><td>2B</td><td>K4</td>
</tr>
<tr>
<td>8 Ans</td>
<td>
1. The Concept of Feature Selection (2 Marks)
Unlike feature extraction (which creates new features), feature selection maintains the original form of the data but removes redundant or irrelevant attributes.

2. Analysis of Feature Selection Methods (6 Marks)
A. Filter Methods
These methods use statistical measures to assign a scoring to each feature. They are "model-agnostic," meaning they look at the data without actually running a machine learning algorithm.
B. Wrapper Methods
These methods treat the selection process as a search problem. They train a model on different subsets of features and select the subset that produces the best performance.
C. Embedded Methods
These methods perform feature selection during the model training process. The selection is "embedded" in the algorithm's logic.
3. Impact on Performance and Complexity (5 Marks)
Feature selection plays a dual role in optimizing the machine learning lifecycle:
</td>
<td>CO2</td><td>2B</td><td>K4</td>
</tr>
</table>

	I. Improving Model Performance ● Reduction of Overfitting: ● Accuracy Boost:. II. Reducing Model Complexity ● Interpretability: ● Computational Efficiency			

Part- C

<table>
<tr><th>Q.No</th><th>Questions</th><th>CO's</th><th>Unit</th><th>Bloom's Level</th></tr>
<tr><td>1.</td><td>A retail company collects customer data from online transactions, mobile apps, and third-party vendors. The dataset contains missing values, inconsistent formats, and duplicate records.
a) Evaluate the suitability of different data sources and data collection methods used by the company.
b) Critically assess the data quality issues present in the dataset and justify appropriate data cleaning techniques to improve data reliability.
<table>
<tr><th>Customer_ID</th><th>Name</th><th>Source</th><th>Age</th><th>Gender</th><th>Purchase_Amount (₹)</th><th>Last_Login_Date</th><th>City</th></tr>
<tr><td>C101</td><td>Anil Kumar</td><td>Web Logs</td><td>28</td><td>M</td><td>4500</td><td>12-08-2025</td><td>Chenn</td></tr>
<tr><td>C102</td><td>Priya S</td><td>Mobile App</td><td></td><td>F</td><td>3200</td><td>10-08-2025</td><td>Bengaru</td></tr>
<tr><td>C103</td><td>Rahul Mehta</td><td>Survey</td><td>35</td><td>Male</td><td></td><td>10-08-2025</td><td>Mumb</td></tr>
<tr><td>C101</td><td>Anil Kumar</td><td>Mobile App</td><td>28</td><td>M</td><td>4500</td><td>08-12-2025</td><td>Chenn</td></tr>
<tr><td>C104</td><td>Sneha R</td><td>Third Party</td><td>42</td><td>F</td><td>7800</td><td>09-08-2025</td><td>Delh</td></tr>
</table></td><td>CO2</td><td>2 C</td><td>K5</td></tr>
</table>

	C105	John Paul	Web Logs	31		5200		Hyderabad			
	C106	Ramesh K	Third Party	29	M	5000	11-08-2025	chenn		
1 Ans	**a) Evaluation of Data Sources and Data Collection Methods (7 Marks)** **1. Web Logs** ● records and lacks detailed demographic attributes. **2. Mobile Application Data** **3. Survey Data** **4. Third-Party Vendor Data** **b) Data Quality Issues and Data Cleaning Techniques (8 Marks)** **1. Missing Values** **Observed Issues** **Cleaning Techniques** **2. Inconsistent Data Formats** **Observed Issues** **Cleaning Techniques** ● Standardize gender as Male / Female. ● Convert city names to consistent case. ● Normalize dates to ISO format (YYYY-MM-DD). **3. Duplicate Records** **Observed Issue** ● Deduplication using Customer_ID. ● Retain the most recent or most complete record. ● Merge data from multiple sources when necessary. **4. Source-Level Inconsistencies** **Observed Issue** ● Same customer information varies across data sources. **Cleaning Techniques** ● Master Data Management (MDM). ● Define source priority (Mobile App > Web Logs > Survey). **Data Cleaning Workflow Diagram** Data Collection (Web, Mobile App, Survey, Third Party) ↓ Data Integration ↓ Data Profiling (Identify missing values, duplicates, inconsistencies) ↓ Data Cleaning • Missing Value Imputation • Format Standardization • Deduplication ↓								CO2	2 C	K5

	Data Validation (Accuracy, Consistency, Completeness) ↓ Clean & Reliable Dataset ↓ Analytics & Decision Making			
2.	A healthcare dataset contains noisy sensor readings and several extreme values that affect predictive model accuracy. a) Evaluate different noise and outlier detection techniques applicable to this dataset. b) Justify the choice of suitable smoothing and data transformation methods to enhance data quality and model performance. **Patient_ID** | **Heart_Rate (bpm)** | **Blood_Pressure (mmHg)** | **Body_Temp (°C)** | **Oxygen_Level (%)** P001 | 72 | 120 | 36.8 | 98 P002 | 75 | 118 | 36.7 | 97 P003 | 180 | 200 | 39.5 | 85 P004 | 70 | 122 | 36.9 | 99 P005 | 74 | 119 | 36.6 | 98 P006 | 35 | 60 | 34 | 88 P007 | 73 | 121 | 36.8 | 97 P008 | 250 | 210 | 40.2 | 82	CO2	2 C	K5
2 Ans	**a) Noise and Outlier Detection Techniques (7 Marks)** 1. Z-Score Method ● Measures deviation of values from the mean. ● Values with $\|Z\| > 3$ are treated as outliers. 2. Interquartile Range (IQR) Method ● Identifies outliers outside: o $Q1 - 1.5 \times IQR$ o $Q3 + 1.5 \times IQR$ 3. Domain-Based Threshold Detection ● Uses medical knowledge to define valid ranges: o Heart rate: 60–100 bpm o Body temperature: 36–37.5 °C o Oxygen level: >90% 4. Machine Learning-Based Techniques ● Examples: Isolation Forest, DBSCAN. ● Detects anomalies based on data density and isolation. **b) Smoothing and Data Transformation Methods (8 Marks)** 1. Smoothing Techniques a) Moving Average Smoothing ● Averages neighboring readings to reduce noise. b) Median Filtering ● Replaces noisy values with the median of nearby readings.	CO2	2 C	K5

	2. Data Transformation Techniques a) Log Transformation ● Compresses extreme values. b) Normalization (Min–Max / Z-Score) ● Scales all attributes to a common range. c) Winsorization ● Caps extreme values at predefined percentile limits. Noise & Outlier Handling Workflow Diagram (Exam-Friendly) Healthcare Sensor Data ↓ Data Profiling (Detect noise & extreme values) ↓ Outlier Detection • Z-Score • IQR • Domain Thresholds • ML-based Methods ↓ Noise Reduction • Moving Average • Median Filtering ↓ Data Transformation • Log Scaling • Normalization • Winsorization ↓ Clean & Reliable Dataset ↓ Improved Predictive Model Performance			

Unit- 3

Part A

Q.No	Questions	CO's	Unit	Bloom's Level
1.	Recall descriptive statistics?	CO3	3A	K1
1 Ans	Descriptive statistics are methods used to summarize, organize, and present data in a meaningful way using measures such as mean, median, mode, tables, and graphs.	CO3	3A	K1
2.	Name the different types of descriptive statistics.	CO3	3A	K1
2 Ans	The main types of descriptive statistics are: ● Measures of central tendency ● Measures of dispersion	CO3	3A	K1
3.	State the relation between mean, median, and mode.	CO3	3A	K1
3 Ans	$Mode = 3(Median) - 2(Mean)$	CO3	3A	K1
4.	The marks obtained by 7 students are: 5, 8, 5, 10, 7, 8, 5. Find mean, median, and mode.	CO3	3A	K3
4 Ans	Data (sorted): 5, 5, 5, 7, 8, 8, 10	CO3	3A	K3

	• Mean = Mean = $\frac{48}{7} \approx 6.86$ • Median = 7 • Mode = 5			
5.	A dataset has a mean of 50 and standard deviation of 10. Identify the range that covers approximately 68% of the data.	CO3	3A	K3
5 Ans	Mean=50 Standard deviation =10 $Range = \mu \pm \sigma = 50 \pm 10$ Hence, the range is 40 to 60.	CO3	3A	K3
6.	Define probability in the context of data science.	CO3	3A	K1
6 Ans	Probability is a numerical measure that quantifies the likelihood of occurrence of an event, with values ranging from 0 to 1.	CO3	3A	K1
7.	Determine whether the given data is best represented by a discrete or continuous random variable: "Number of students present in a class".	CO3	3A	K3
7 Ans	It is a discrete random variable, since the number of students can take only whole numbers.	CO3	3A	K3
8.	Write the mean and variance of Poisson distribution.	CO3	3A	K1
8 Ans	For a Poisson distribution with parameter : • Mean(E[X]) =λ • Variance(var[X]) =λ	CO3	3A	K1
9.	State two properties of normal distribution.	CO3	3A	K1
9 Ans	• It is symmetrical about the mean. • Mean = Median = Mode.	CO3	3A	K1
10.	Suppose a basketball player has a free-throw success rate of 70% (p = 0.7). If the player takes 10 shots (n = 10), we want to find the mean and variance of the number of successful shots.	CO3	3A	K3
10 Ans	μ=n*p Mean=7,Variance=2.1	CO3	3A	K2
11.	Define Poisson distribution.	CO3	3A	K1
11 Ans	A Poisson distribution models the probability of a given number of events occurring in a fixed interval of time or space when events occur independently.	CO3	3A	K1
12.	Identify the type of distribution suitable when counting the number of arrivals in a fixed time interval.	CO3	3A	K3
12 Ans	The suitable distribution for counting the number of arrivals in a fixed time interval is the Poisson distribution.	CO3	3A	K3
13.	List any two sampling methods used in data science.	CO3	3A	K1
13 Ans	Two sampling methods are: • Simple Random Sampling • Stratified Sampling	CO3	3A	K1
14.	State the Central Limit Theorem.	CO3	3A	K2
14 Ans	The Central Limit Theorem states that the sampling distribution of the sample mean approaches a normal distribution as the sample size increases, regardless of the population distribution.	CO3	3A	K2
15.	How does correlation differ from covariance?	CO3	3A	K4

15 Ans	• Covariance indicates the direction of the relationship between two variables. • Correlation indicates both the direction and strength of the relationship and is scale-independent.	CO3	3A	K4

Part- B

Q.No	Questions	CO's	Unit	Bloom's Level
1.	Discuss how measures of dispersion complement measures of central tendency. Use examples to show situations where mean alone is not enough.	CO3	3B	K4
1 Ans	**Measures of Central Tendency(5 Marks)** • Mean – arithmetic average • Median – middle value • Mode – most frequent value **Measures of Dispersion (8 Marks)** • Range – difference between maximum and minimum • Variance – average of squared deviations • Standard Deviation – square root of variance Why Mean Alone Is Not Enough Example 1: Two datasets • Dataset A: {50, 50, 50, 50, 50} • Dataset B: {10, 30, 50, 70, 90} Both have mean = 50, but: • Dataset A → no variation (low dispersion) • Dataset B → high variation (high dispersion) Mean alone does not show variability. Diagram: Mean with Low vs High Dispersion Low Dispersion High Dispersion |----|----|----| |-----|----|------| 50 50 50 10 50 90 **Importance of Dispersion** • Shows consistency of data • Helps compare datasets with same mean • Indicates risk and variability Measures of dispersion complement central tendency by explaining how data values are distributed around the average. Without dispersion, the mean can be misleading.	CO3	3B	K4
2.	Explain descriptive statistics and its significance in data analysis. Illustrate with examples how measures of central tendency and dispersion are used in summarizing data.	CO3	3B	K3
2 Ans	**Definition** Descriptive statistics are methods used to summarize, organize, and present data meaningfully using numerical measures and graphical tools. **Components of Descriptive Statistics (5 Marks)** 1. Measures of Central Tendency • Mean • Median • Mode	CO3		K3

	2. Measures of Dispersion ● Range ● Variance ● Standard Deviation **Significance of Descriptive Statistics** ● Simplifies large datasets ● Helps identify patterns and trends ● Enables comparison between datasets ● Supports decision making **Example Dataset (5 Marks)** Marks: {40, 45, 50, 55, 60} ● Mean = 50 ● Median = 50 ● Range = 60 − 40 = 20 **Graphical Representation (3 Marks)** Marks Frequency 40 * 45 ** 50 *** 55 ** 60 * **Use in Data Analysis** ● Mean → average performance ● Median → typical value ● Standard deviation → variation in performance		3B	
3.	The average number of calls received by a customer care center per minute is 3. Using the Poisson distribution, find the probability that exactly two calls are received in a given minute. Determine the probability that at most four calls are received in a minute and justify the use of the Poisson distribution for this scenario.	CO3	3B	K5
3 Ans	**Given** Average number of calls per minute = λ = 3 Poisson probability formula: **(a) Probability of Exactly Two Calls(5 Marks)** $P(X = 2) = \frac{e^{-3}3^2}{2!} = \frac{e^{-3}\times 9}{2} \approx \frac{0.0498\times 9}{2} = 0.224 P(X = 2$ **(b) Probability of At Most Four Calls(5 Marks)** $P(X \leq 4) = P(0) + P(1) + P(2) + P(3) + P(4)$ $= e^{-3}(1 + 3 + 4.5 + 4.5 + 3.375)$ $= e^{-3}(16.375) \approx 0.815$ $P(X \leq 4) \approx 0.815$ **Justification for Using Poisson Distribution(3 marks)** ● Events occur independently ● Fixed time interval (1 minute)	CO3	3B	K5

	• Known average rate (λ = 3) • Suitable for counting arrivals **Poisson Process Diagram** Time (1 min) |------|------|------| |Call |Call | Call |			
4.	The heights of students in a college are normally distributed with mean 165 cm and standard deviation 5 cm. i) Find the probability that a student's height is between 160 cm and 170 cm. ii) Explain why the normal distribution is used for modelling height.	CO3	3B	K5
4 Ans	**Given Data** • Mean height, cm • Standard deviation, cm • Heights are normally distributed **We are required to find (7 marks)** **Step 1:** Convert to Standard Normal Variable (Z-score) The Z-score formula is: **The Z-score formula is:** $Z = \frac{X-\mu}{\sigma}$ **For $X = 160$:** $Z_1 = \frac{160-165}{5} = -1$ **For $X = 170$:** $Z_2 = \frac{170-165}{5} = 1$ **Thus,** $P(160 < X < 170) = P(-1 < Z < 1)$ **Step 2:** Use Standard Normal Distribution Table From standard normal tables: $P(-1 < Z < 1) = 0.6826$ Final Answer (Probability) $P(160 < X < 170) = 0.6826$ **Interpretation (6 marks)** There is approximately a 68.26% probability that a randomly selected student's height lies between 160 cm and 170 cm. Why Normal Distribution Is Used to Model Height 1. Biological factors: 2. Symmetry: 3. Empirical evidence: 4. Observed height data in large populations closely follows a bell-shaped curve.	CO3	3B	K5

<table>
<tr><td></td><td>5. Statistical properties:
Importance in Data Analysis
● Helps estimate probabilities for height ranges
● Useful in health studies, education, and ergonomics
● Supports prediction and comparison across populations</td><td></td><td></td><td></td></tr>
<tr><td>5.</td><td>Discuss the difference between discrete and continuous probability distributions. Evaluate why certain phenomena are better modelled using continuous distributions rather than discrete ones.</td><td>CO3</td><td>3B</td><td>K5</td></tr>
<tr><td>5
Ans</td><td>Discrete Probability Distribution(5 marks)
Definition
A discrete probability distribution is used when a random variable takes countable and distinct values.
Characteristics
● Values are finite or countably infinite
● Probability is assigned to each exact value
● Represented using Probability Mass Function (PMF)
● Sum of probabilities equals 1
Examples
● Number of students in a class
● Number of calls received per hour
● Number of defective items

Continuous Probability Distribution (5 marks)
Definition
A continuous probability distribution is used when a random variable can take any value within a given range.
Characteristics
● Infinite possible values
● Probability of a single value is zero
● Probability is found over an interval
● Represented using Probability Density Function (PDF)
Examples
● Height of students
● Weight of a person
● Time taken to complete a task

Comparison Between Discrete and Continuous Distributions(3 marks)
<table>
<tr><th>Aspect</th><th>Discrete Distribution</th><th>Continuous Distribution</th></tr>
<tr><td>Nature of values</td><td>Countable</td><td>Uncountable</td></tr>
<tr><td>Probability</td><td>Assigned to exact values</td><td>Assigned to intervals</td></tr>
<tr><td>Function used</td><td>PMF</td><td>PDF</td></tr>
<tr><td>Graph</td><td>Bar graph</td><td>Smooth curve</td></tr>
</table>
</td><td>CO3</td><td>3B</td><td>K5</td></tr>
</table>

	Example | Number of arrivals | Height, weight				
6.	Analyze the concept of stratified and cluster sampling. Compare their advantages and disadvantages. In what situations would each method be preferred?		CO3	3B	K4
6 Ans	**Stratified Sampling(5 marks)** **Working** ● Divide population into strata based on characteristics (age, gender, income) ● Randomly select samples from each stratum ● Combine all selected samplesAdvantages of Stratified Sampling ● Ensures representation of all groups ● Higher accuracy and precision ● Reduces sampling error **Disadvantages of Stratified Sampling** ● Requires detailed population information ● Complex to implement ● Time-consuming **Cluster Sampling (5 marks)** Working ● Divide population into clusters (schools, cities) ● Randomly select clustersAdvantages of Cluster Sampling ● Cost-effective ● Easy to implement ● Suitable for large geographical areas Disadvantages of Cluster Sampling ● Higher sampling error ● Less accurate than stratified sampling ● Possible cluster bias **Situations Where Each Method Is Preferred(3 marks)** **Stratified Sampling Preferred When:** ● Population has distinct subgroups ● High accuracy is required ● Size of strata varies significantly **Cluster Sampling Preferred When:** ● Population is widely dispersed ● Cost and time constraints exist ● Detailed population data is unavailable Stratified sampling improves accuracy by ensuring proportional representation, while cluster sampling reduces cost and effort. The choice depends on the research objective, population structure, and available resources.		CO3	3B	K4
7.	Illustrate the Central Limit Theorem (CLT). State its assumptions and significance. Illustrate with an example how it is used in estimating population parameters.		CO3	3B	K4
7 Ans	**Mathematical Representation(7 marks)** If: ● Population mean = μ ● Population standard deviation = σ		CO3	3B	K4

	● Sample size = n Then the sampling distribution of the mean has: $Mean = \mu Standard\ Error = \frac{\sigma}{\sqrt{n}}$ **Assumptions of CLT** 1. Samples are randomly selected 2. Observations are independent 3. Sample size is sufficiently large (usually $n \geq 30$) 4. Population has a finite mean and variance Significance of Central Limit Theorem ● Allows use of normal distribution for inference ● Enables estimation of population parameters ● Forms basis for confidence intervals and hypothesis testing ● Simplifies analysis of non-normal populations **Illustration with Example(6 marks)** **Example Scenario** A population of daily incomes is skewed. ● Population mean = ₹1000 ● Population standard deviation = ₹300 ● Sample size = 36 Using CLT: $Standard\ Error = \frac{300}{\sqrt{36}} = 50$ The sampling distribution of the sample mean is approximately **normal**, even though the population is skewed. **Diagram: CLT Concept** Population Distribution Sampling Distribution of Mean (Skewed) (Normal) **Use in Estimating Population Parameters** Using the normal sampling distribution: ● Probability estimates can be computed ● Confidence intervals for population mean can be constructed ● Decision making becomes statistically reliable			
8.	The marks obtained by 5 students in Mathematics (X) and Physics (Y) are as follows: X: 65, 70, 75, 80, 85 Y: 70, 72, 78, 75, 80 i) Calculate the covariance between X and Y. ii) Calculate the correlation coefficient between X and Y. iii) Comment on the strength and direction of the relationship.	CO3	3B	K4
	Step 1: Calculate Mean of X and Y(5 marks) $\bar{X} = \frac{65+70+75+80+85}{5} = \frac{375}{5} = 75 \bar{Y} = \frac{70+72+78+75+80}{5} =$			

8 Ans	Formula $Cov(X,Y) = \frac{\sum(X-\bar{X})(Y-\bar{Y})}{n}$ Calculation $Cov(X,Y) = \frac{115}{5} = 23$ **(ii) Correlation Coefficient Between X and Y (5 marks)** Formula $r = \frac{\sum(X-\bar{X})(Y-\bar{Y})}{\sqrt{\sum(X-\bar{X})^2 \sum(Y-\bar{Y})^2}}$ Calculation $r = \frac{115}{\sqrt{250\times68}} r = \frac{115}{\sqrt{17000}} = \frac{115}{130.38} r \approx 0.88$ **(iii) Comment on Strength and Direction(3 marks)** ● The covariance is positive, indicating a direct relationship ● The correlation coefficient (r ≈ 0.88) indicates a strong positive correlation ● As Mathematics marks increase, Physics marks also tend to increase	CO3	3B	K4

Part- C

Q.No	Questions	CO's	Unit	Bloom's Level
1.	a) The following data represents the number of customer visits to a store over 12 days: 22, 18, 25, 30, 20, 27, 24, 19, 26, 28, 21, 29. Using suitable descriptive statistical measures, analyze the data and comment on the central tendency, dispersion and overall variability of the dataset. b) A manufacturing process produces items where the number of defective items per day follows a Poisson distribution with a mean of 3. Determine the probability that at most four defective items are produced in a day.Also briefly justify why the Poisson distribution is appropriate in this situation.	CO3	3C	K5
1 Ans	**(a) Descriptive Statistical Analysis of Customer Visits (7 marks)** Given Data (12 days) 22, 18, 25, 30, 20, 27, 24, 19, 26, 28, 21, 29 Step 1: Arrange Data in Ascending Order 18, 19, 20, 21, 22, 24, 25, 26, 27, 28, 29, 30 Step 2: Measures of Central Tendency Mean $\bar{X} = \frac{\sum X}{n} = \frac{289}{12} \approx 24.08$ Median For even number of observations:	CO3	3C	K5

	$Median = \frac{6^{th}+7^{th}}{2} = \frac{24+25}{2} = 24.5$ Mode There is no mode, as all values occur only once. Step 3: Measures of Dispersion Range $Range = Max - Min = 30 - 18 = 12$ Standard Deviation (Approximate) $\sigma \approx 4$ **(b) Poisson Distribution Problem (8 marks)** Given ● Mean number of defects per day, $\lambda = 3$ Poisson probability formula: $P(X = x) = \frac{e^{-\lambda}\lambda^{x}}{x!}$ Required $P(X \leq 4) = P(0) + P(1) + P(2) + P(3) + P(4)$ Step 1: Calculate Individual Probabilities $P(0) = e^{-3} = 0.0498 P(1) = 3e^{-3} = 0.1494 P(2)$ Step 2: Total Probability $P(X \leq 4) = 0.0498 + 0.1494 + 0.2240 + 0.2240$ Justification for Using Poisson Distribution ● Counts number of events (defects) ● Occurs in a fixed time interval (per day) ● Events occur independently ● Average rate (λ) is known and constant			
2.	Compare Binomial distribution and Poisson distribution. Analyze the specific conditions (in terms of n and p) under which the Binomial distribution can be approximated by the Poisson distribution. Provide one real-world application for each.	CO3	3C	K4
2 Ans	**Binomial Distribution (7 marks)** $P(X = x) = \left(\frac{n}{x}\right)p^{x}(1 - p)^{n-x}$ **Definition** The binomial distribution gives the probability of obtaining exactly *x* successes in *n* independent Bernoulli trials, each with success probability *p*. Probability Mass Function **Real-World Application** ● Number of defective items in a batch of fixed size ● Number of heads in *n* coin tosses	CO3	3C	K4

Poisson Distribution
Definition
The Poisson distribution gives the probability of x events occurring in a fixed time or space interval, when events occur independently at a constant average rate.

Probability Mass Function

$$P(X = x) = \frac{e^{-\lambda}\lambda^{x}}{x!}$$

Real-World Application
- Number of calls arriving at a call center per minute
- Number of accidents at a junction per day

Comparison Between Binomial and Poisson Distributions

Aspect	**Binomial Distribution**	**Poisson Distribution**
Type of experiment	Fixed number of trials	Fixed interval of time/space
Number of trials	Finite (n)	Infinite/undefined
Probability of success	Constant (p)	Not defined
Parameter	n, p	Mean rate (λ)
Mean	Np	Λ
Variance	$np(1-p)$	Λ
Nature of events	Two outcomes per trial	Count of events
Example	Coin tossing	Call arrivals

Poisson Approximation to Binomial Distribution
Conditions for Approximation (8 marks)
The binomial distribution can be approximated by the Poisson distribution when:
1. The number of trials n is large
2. The probability of success p is very small
3. The product $\lambda = np$ is finite and moderate

Explanation
When n increases and p decreases such that np remains constant, the binomial probability formula simplifies and approaches the Poisson probability formula. This approximation reduces computational complexity.

Mathematical Relation
If:

$$\lambda = np$$

	Then $\left(\frac{n}{x}\right)p^{x}(1-p)^{n-x} \approx \frac{e^{-\lambda}\lambda^{x}}{x!}$ **Example of Approximation** ● Number of trials: n = 1000 ● Probability of success: p = 0.002 ● Mean: λ = np = 2 Here, Poisson distribution is a good approximation to the binomial distribution.			

Unit- 4
Part A

Q.No	Questions	CO's	Unit	Bloom's Level
1.	Define Machine Learning with suitable example.	CO4	4A	K1
1 Ans	**Machine Learning (ML)** is the field of study that gives computers the ability to learn without being explicitly programmed. It involves developing statistical algorithms that can generalize from a given dataset to perform tasks on new, unseen data. Example: Email Spam Filter Medical Imaging Banking Self-Driving Cars	CO4	4A	K1
2.	Recall the niche of machine learning?	CO4	4A	K1
2 Ans	The niche of Machine Learning (ML) lies in its ability to **bridge the gap between raw data and automated decision-making** without explicit programming. Its specific "place" or function is defined by two key roles: 1 .Pattern Discovery in Big Data 2. Adaptive Intelligence	CO4	4A	K1
3.	List differences between supervised and unsupervised learning.	CO4	4A	K2
3 Ans	<table><tr><td>Feature</td><td>Supervised Learning</td><td>Unsupervised Learning</td></tr><tr><td>Data Type</td><td>Uses labeled data (input-output pairs).</td><td>Uses unlabeled data (raw input only).</td></tr><tr><td>Learning Goal</td><td>To predict an outcome or classify new data.</td><td>To find hidden patterns or structures in data.</td></tr></table>	CO4	4A	K2

	Feedback	Direct feedback (the model is told the "correct" answer).	No feedback (t model learns o its own).			
	Algorithms	Regression, Classification, Decision Trees, SVM.	Clustering (K-Means), Association, Dimensionality Reduction.			
	Complexity	Computationally simpler to evaluate.	Computational more complex and subjective.			
	Common Tasks	Spam detection, price prediction, diagnosis.	Customer segmentation, anomaly detection.[2]			
4.	Compare Overfitting and Underfitting.			CO4	4A	K2
4 Ans	Overfitting occurs when a model learns the noise in training data too well and fails on new data; Underfitting occurs when the model is too simple to learn the underlying trend even in training data			CO4	4A	K2
5.	When should you use Logistic Regression instead of Linear Regression?			CO4	4A	K2
5 Ans	Use Logistic Regression when the target variable is categorical (e.g., Yes/No), and Linear Regression when the target variable is continuous (e.g., Price/Weight).			CO4	4A	K2
6.	Outline the main steps of the K-means algorithm			CO4	4A	K3
6 Ans	1.Initialize K centroids randomly. 2. Assign each data point to the nearest centroid. 3. Update centroids by calculating the mean of assigned points. 4. Repeat until convergence.			CO4	4A	K3
7.	In a model with high Variance but low Bias, which problem is the model likely facing?			CO4	4A	K3
7 Ans	The model is likely facing Overfitting. High variance means the model is highly sensitive to small fluctuations in the training set, leading to poor generalization.			CO4	4A	K3
8.	Calculate the Mean of a Binomial Distribution if n=50 and p=0.2			CO4	4A	K3
8 Ans	$\mu=n*p$ Mean= 10			CO4	4A	K3
9.	What is cross-validation?			CO4	4A	K1
9 Ans	In machine learning, Cross-Validation is a resampling technique used to evaluate how a model will perform on an independent, unseen dataset. It is primarily used to flag problems like overfitting and			CO4	4A	K1

<table>
<tr><td></td><td>to ensure the model generalizes well rather than just memorizing the training data.
Instead of splitting the data once (like a simple Train/Test split), cross-validation splits the data multiple times in different ways and averages the results</td><td></td><td></td><td></td></tr>
<tr><td>10.</td><td>How can you use "Pruning" to handle Overfitting in Decision Trees?</td><td>CO4</td><td>4A</td><td>K2</td></tr>
<tr><td>10 Ans</td><td>Pruning involves removing sections of the tree that provide little power to classify instances. This reduces the complexity of the final classifier, which improves predictive accuracy by reducing overfitting.</td><td>CO4</td><td>4A</td><td>K2</td></tr>
<tr><td>11.</td><td>If a Linear Regression model has a high training error and a high testing error, what is the problem and how do you fix it?</td><td>CO4</td><td>4A</td><td>K3</td></tr>
<tr><td>11 Ans</td><td>The problem is Underfitting. It can be fixed by increasing the model complexity (e.g., adding more features) or by using a more sophisticated algorithm that can capture non-linear trends.</td><td>CO4</td><td>4A</td><td>K3</td></tr>
<tr><td>12.</td><td>Why is K-means considered an "Unsupervised" algorithm?</td><td>CO4</td><td>4A</td><td>K2</td></tr>
<tr><td>12 Ans</td><td>It is unsupervised because it works on unlabeled data. It does not have a "teacher" or target variable to learn from; instead, it finds inherent structures or groupings based on the similarity of data points</td><td>CO4</td><td>4A</td><td>K2</td></tr>
<tr><td>13.</td><td>List the advantages and disadvantages of decision tree.</td><td>CO4</td><td>4A</td><td>K1</td></tr>
<tr><td>13 Ans</td><td><table><tr><th>Feature</th><th>Advantage</th><th>Disadvantage</th></tr><tr><td>Logic</td><td>Transparent & easy to explain</td><td>Can be "greedy" and sub-optimal</td></tr><tr><td>Complexity</td><td>Handles non-linear data well</td><td>Prone to overfitting (memorizing noise)</td></tr><tr><td>Data Prep</td><td>No scaling or encoding needed</td><td>Sensitive to sma data variations</td></tr><tr><td>Output</td><td>Easy to visualize</td><td>Not smooth (piecewise constant)</td></tr></table></td><td>CO4</td><td>4A</td><td>K1</td></tr>
<tr><td>14.</td><td>Given a confusion matrix where TP=80, FP=10, FN=10, and TN=100, calculate the Precision.</td><td>CO4</td><td>4A</td><td>K3</td></tr>
<tr><td>14 Ans</td><td>$Precision = TP + FP/$ TP
Ans:0.89 or 89%</td><td>CO4</td><td>4A</td><td>K3</td></tr>
<tr><td>15.</td><td>How can overfitting be prevented?</td><td>CO4</td><td>4A</td><td>K2</td></tr>
</table>

15 Ans	Preventing **overfitting** is about forcing the model to focus on the "signal" (the true pattern) rather than the "noise" (random fluctuations in your specific training data). ● Data-Based Techniques ● Model-Based Techniques (Regularization) ● Training Process Techniques	CO4	4A	K2

Part- B

Q.No	Questions	CO's	Unit	Bloom's Level
1.	Analyse various learning techniques involved in unsupervised learning?	CO4	4B	K4
1 Ans	**Unsupervised Learning** **1. Definition and Concept(3 Marks)** Unsupervised Learning is a type of machine learning where the algorithm learns from a dataset that does not have labeled responses (no target variable). The goal is to model the underlying structure or distribution in the data to learn more about it. Unlike supervised learning, there is no "teacher" or "correct answer" to guide the process. **2. Primary Learning Techniques(4 Marks)** Unsupervised learning is generally categorized into four major techniques: **A. Clustering (Grouping Data)** Clustering is the process of partitioning a dataset into groups (clusters) such that data points in the same group are similar, while points in different groups are distinct. ● **K-Means Clustering:** ● **Hierarchical Clustering** ● **DBSCAN** **B. Dimensionality Reduction (Simplifying Data)** This technique reduces the number of input variables in a dataset while preserving as much information as possible. It is crucial for visualizing high-dimensional data and removing noise. ● **Principal Component Analysis (PCA):** A **t-SNE:** **C. Association Rule Learning (Finding Relationships)** This discovers "if-then" relationships between variables in large databases. It is widely used in transactional data analysis. ● **Apriori Algorithm** ● **Metrics** **D. Anomaly Detection (Outlier Analysis)** This identifies rare items or events that differ significantly from the majority of the data. ● **Isolation Forest:** **3. Comparison: Supervised vs. Unsupervised(2 Marks)**	CO4	4B	K4

Feature	**Supervised Learning**	**Unsupervised Learning**
Data	Labeled (Input + Output)	Unlabeled (Input only)

<table>
<tr><td></td><td>
<table>
<tr><td>Goal</td><td>Predict outcomes</td><td>Discover hidden patterns</td></tr>
<tr><td>Complexity</td><td>Simple (direct feedback)</td><td>Complex (no feedback)</td></tr>
<tr><td>Accuracy</td><td>High (can be measured)</td><td>Subjective (hard to validate)</td></tr>
</table>
4. Applications(2 Marks)
i) Customer Segmentation:
ii) Image Compression
iii) Medical ImagingFraud Detectio
5. Challenges(2 Marks)
● Evaluation Difficulty
● Computational Cost
● Scalability
</td><td></td><td></td><td></td></tr>
<tr><td>2.</td><td>Why do we use Regression Analysis? Classify the types of Regression with an example</td><td>CO4</td><td>4B</td><td>K4</td></tr>
<tr><td>2 Ans</td><td>
Why Do We Use Regression Analysis? (2 marks)
Prediction and Forecasting
Determining Relationship Strength
Trend Analysis
Error Correction.
Types of Regression (8 marks)
1. Simple Linear Regression
2. Multiple Linear Regression
3. Logistic Regression
4. Polynomial Regression
5. Ridge and Lasso Regression (Regularization)
6. Support Vector Regression (SVR).
Comparison Table (3 marks)
<table>
<tr><td>Regression Type</td><td>Dependent Variable</td><td>Relationship Type</td><td>Key Feature</td></tr>
<tr><td>Simple Linear</td><td>Continuous</td><td>Linear</td><td>1 Independent Variable</td></tr>
<tr><td>Multiple Linear</td><td>Continuous</td><td>Linear</td><td>2+ Independent Variables</td></tr>
<tr><td>Logistic</td><td>Categorical (Binary)</td><td>Non-linear (Sigmoid)</td><td>Predicts Probabilities</td></tr>
<tr><td>Polynomial</td><td>Continuous</td><td>Curvilinear</td><td>Uses powers of x</td></tr>
<tr><td>Lasso</td><td>Continuous</td><td>Linear</td><td>Performs Feature Selection</td></tr>
</table>
</td><td>CO4</td><td>4B</td><td>K2</td></tr>
<tr><td>3.</td><td>Illustrate how Logistic regression used to solve the classification problems</td><td>CO4</td><td>4B</td><td>K3</td></tr>
<tr><td>3 Ans</td><td>Solving Classification Problems with Logistic Regression
1. The Core Mechanism: The Sigmoid Function (2 marks)
The fundamental difference between regression and classification lies in how the output is handled. Logistic</td><td>CO4</td><td>4B</td><td>K3</td></tr>
</table>

<table>
<tr>
<td></td>
<td>Regression uses the Sigmoid (or Logistic) Function to map any real-valued number into a value between 0 and 1
2. How Classification is Achieved (3 marks)
Logistic Regression solves classification problems through a three-step process:
A. Calculating the Probability
B. The Decision Boundary
C. The Cost Function (Log Loss)
3. Types of Logistic Regression (4 marks)
While typically used for binary tasks, it can be extended:
● Binary Logistic Regression: Only two possible outcomes (e.g., Cancer vs. No Cancer).
● Multinomial Logistic Regression: Three or more unordered categories (e.g., Predicting if an image is a Cat, Dog, or Bird).
● Ordinal Logistic Regression: Three or more categories with a specific order (e.g., Rating a movie from 1 to 5 stars).
4. Advantages and Limitations (2 marks)
<table>
<tr><th>Advantages</th><th>Limitations</th></tr>
<tr><td>Highly efficient and easy to implement.</td><td>Assumes a linear relationship between features and log-odds.</td></tr>
<tr><td>Provides a probability score, not just a hard label.</td><td>Struggles with complex, non-linear patterns (where trees or neural nets excel).</td></tr>
<tr><td>Less prone to overfitting in low-dimensional spaces.</td><td>Requires a large sample size for stable results.</td></tr>
</table>
5. Practical Example: Medical Diagnosis (2 marks)
Suppose a doctor wants to predict if a tumour is Malignant (1) or Benign (0) based on its Size.
● The model learns that as size increases, the probability of being malignant increases.
● The Sigmoid curve will show a sharp rise at a specific size threshold.</td>
<td></td><td></td><td></td>
</tr>
<tr>
<td>4.</td>
<td>Compare and contrast Simple Linear Regression and Multiple Linear Regression. Provide the mathematical model for each and explain the role of the error term . Discuss the concept of Decision Trees for classification. Explain the significance of Root Nodes, Internal Nodes, and Leaf Nodes in the decision-making process.</td>
<td>CO4</td><td>4B</td><td>K3</td>
</tr>
<tr>
<td>4 Ans</td>
<td>1.Linear Regression Comparison (4 Marks)
<table>
<tr><th>Feature</th><th>Simple Linear Regression (SLR)</th><th>Multiple Linear Regression (MLR)</th></tr>
<tr><td>Input Variables</td><td>Single independent variable (X).</td><td>Multiple independent variables ($X_1, X_2, \dots, X_n$).</td></tr>
</table>
</td>
<td>CO4</td><td>4B</td><td>K3</td>
</tr>
</table>

	<table><tr><td>Geometry</td><td>A straight line in a 2D plane.</td><td>A hyperplane in a multi-dimensional space.</td></tr><tr><td>Complexity</td><td>Low; easy to visualize and calculate.</td><td>High; requires matrix algebra and handles feature interactions.</td></tr></table> 2. The Mathematical Models (2 marks) 3. Decision Trees for Classification (7 Marks) **A. Root Node (The Starting Point)** ● **Significance:** It represents the entire population or dataset. It is the first attribute chosen to split the data. ● **Decision Power:** The feature selected for the Root Node is the most important predictor, identified using metrics like **Information Gain** or **Gini Impurity**. **B. Internal Nodes (Decision Nodes)** ● **Significance:** These nodes lie between the root and the leaves. Each node represents a "test" or "question" on a specific attribute (e.g., "Is age > 30?"). ● **Decision Power:** They act as filters that further narrow down the data groups to increase the "purity" of the resulting subsets. **C. Leaf Nodes (Terminal Nodes)** ● **Significance:** These are the end points of the tree. They do not split any further. ● **Decision Power:** Each leaf node represents a **class label** (the final prediction). A data point travels from the root through internal nodes until it hits a leaf, which determines its classification.			
5.	A bank wants to predict whether a customer will default on a loan (Yes/No) based on their credit score and age. Analyze the suitability of Logistic Regression versus K-means Clustering for this specific task. Which category of learning does each belong to? .Examine the impact of "Feature Selection" on the performance of a Multiple Linear Regression model. How does including irrelevant features contribute to the Overfitting phenomenon?	CO4	4B	K4
5 Ans	**1.Categorization (2 Marks)** ● **Logistic Regression:** Belongs to **Supervised Learning**. It requires a labeled dataset where the historical outcome (Default: Yes/No) is known. ● **K-means Clustering:** Belongs to **Unsupervised Learning**. It works on unlabeled data to find hidden patterns or groupings based on similarity.	CO4	4B	K4

	2. Suitability Analysis (5 Marks) ● **Logistic Regression (Highly Suitable):** * Since ● **K-means Clustering (Less Suitable):** o K-means would group customers into "clusters" based on proximity in age and credit score, but it **cannot label** them as "Defaulters" without human intervention. o It does not predict a specific outcome; it only identifies who is *similar* to whom. **Feature Selection and Overfitting** **1. Impact of Feature Selection (3 Marks)** ● **Definition:** Feature selection is the process of keeping only the most relevant variables (e.g., Credit Score) and removing noise (e.g., Customer's middle name). ● **Impact: * Reduced Complexity:** Faster training times and simpler models. **2. Irrelevant Features and Overfitting (3 Marks)** ● **The Overfitting Phenomenon:** When a Multiple Linear Regression model includes too many irrelevant features, it becomes "too flexible." ● **Mechanism:** The model starts to treat random noise in the irrelevant features as meaningful patterns. It "memorizes" the training data perfectly (including the errors). ● **Result:** The model shows **Low Bias** (very accurate on training data) but **High Variance** (fails miserably on new, unseen test data).			
6.	Write K-Means algorithm. Suppose that the data mining task is to cluster the following eight points (with (x, y) representing location) into three clusters. A1(2, 10), A2(2, 5), A3(8, 4), B1(5, 8), B2(7, 5), B3(6, 4), C1(1, 2), C2(4, 9). The distance function is Euclidean distance. Suppose initially we assign A1, B1, and C1 as the center of each cluster, respectively. Use the k-means algorithm to show only i)The three cluster centers after the first round of execution and ii) The final three clusters.	CO4	4B	K4
6 Ans	**K-Means Clustering Algorithm** **Input:** ● Dataset with n objects ● Number of clusters ● Distance measure (Euclidean distance) **Algorithm Steps:(3 marks)** ● Select initial cluster centers randomly. ● Assign each data point to the nearest cluster center using the distance measure. ● Recalculate each cluster center as the mean of all points assigned to it. ● Repeat steps 2 and 3 until cluster centers no longer change. ● Output the final clusters and their centers.	CO4	4B	K4

Given Data Points

Point	Coordinates
A1	(2, 10)
A2	(2, 5)
A3	(8, 4)
B1	(5, 8)
B2	(7, 5)
B3	(6, 4)
C1	(1, 2)
C2	(4, 9)

Number of clusters:
Distance: Euclidean

Initial Cluster Centers

- Cluster 1 center = **A1 (2,10)**
- Cluster 2 center = **B1 (5,8)**
- Cluster 3 center = **C1 (1,2)**

Step 1: First Round – Assign Points to Nearest Center (5 marks)
(Using Euclidean distance)
Cluster assignments after first iteration

- **Cluster 1 (A1):**
  A1(2,10), C2(4,9)
- **Cluster 2 (B1):**
  B1(5,8), A2(2,5)
- **Cluster 3 (C1):**
  C1(1,2), A3(8,4), B2(7,5), B3(6,4)

i) Cluster Centers After First Round
Answer (i)

- Cluster 1 center → **(3, 9.5)**
- Cluster 2 center → **(3.5, 6.5)**
- Cluster 3 center → **(5.5, 3.75)**

Step 2: Reassignment Using New Centers (5 marks)
After recalculating distances and reassigning points, clusters stabilize as follows:
ii) Final Three Clusters
Cluster 1

- A1 (2,10)
- C2 (4,9)
- B1 (5,8)

Cluster 2

- A2 (2,5)

Cluster 3

- A3 (8,4)
- B2 (7,5)
- B3 (6,4)
- C1 (1,2)

Final Answer Summary
(i) Cluster centers after first round

- (3, 9.5)
- (3.5, 6.5)

	● (5.5, 3.75) **(ii) Final clusters** ● **Cluster 1:** A1, B1, C2 ● **Cluster 2:** A2 ● **Cluster 3:** A3, B2, B3, C1			
7.	you have developed a Machine Learning model for "Early Cancer Detection." The model shows 99% Accuracy but only 10% Recall. Evaluate whether this model is successful. Justify your judgment using the concepts of the Confusion Matrix and the cost of False Negatives.Critically evaluate the "Bias-Variance Tradeoff" in Decision Trees. How does the process of "Pruning" act as a tool to find the optimal balance between these two errors?	CO4	4B	K5
7 Ans	**Evaluation of Cancer Detection Model (8 Marks)** **1. Judgment of Success (1 Mark)** ● Evaluation: The model is unsuccessful and clinically unsafe for the intended purpose of early cancer detection. **2. Justification via Confusion Matrix & Metrics (4 Marks)** ● The Accuracy Paradox: 99% accuracy is misleading in medical diagnosis because the dataset is likely highly imbalanced (e.g., only 1% of the population has cancer). If a model predicts "No Cancer" for everyone, it achieves 99% accuracy but fails its purpose. **3. The Cost of False Negatives (3 Marks)** ● Clinical Impact: In cancer detection, a False Positive (wrong alarm) leads to further testing (stressful but safe). However, a False Negative (missing the disease) leads to the patient being sent home without treatment, which is life-threatening. **Bias-Variance Tradeoff & Pruning (5 Marks)** **1. Evaluation of Decision Tree behavior (2 Marks)** ● **The Conflict:** Decision Trees are prone to **High Variance**. Because they split data until every leaf is pure, they "memorize" the training data (including noise), leading to **Overfitting**. **2. Pruning as a Balancing Tool (3 Marks)** ● **Mechanism:** Pruning removes branches that provide little predictive power. ● **Impact on Error:** * It slightly increases **Bias** (the model becomes simpler and doesn't fit the training data perfectly). o It significantly decreases **Variance** (the model becomes more stable and generalizes better to unseen data).	CO4	4B	K5
8.	Classify the concepts of Overfitting and Underfitting with respect to the Bias-Variance Tradeoff. Suggesttwo methods to prevent a model from overfitting. Define the Confusion Matrix. Using it, derive the formulas for	CO4	4B	K4

	Accuracy, Precision, and Recall. Why is Recall more important than Accuracy in a "Disease Detection" system?			
8 Ans	Overfitting, Underfitting, and the Bias-Variance Tradeoff **1. The Concepts (3 Marks)** ● **Underfitting:** Occurs when a model is too simple to capture the underlying structure of the data (e.g., using a linear line for a curved relationship). It performs poorly on both training and test data. ● **Overfitting:** Occurs when a model is too complex and captures the noise/random fluctuations in the training data rather than the actual pattern. It performs excellently on training data but fails on new, unseen data. **2. Bias-Variance Tradeoff (2 Marks)** The total error of a model is the sum of **Bias** and **Variance**. ● High Bias (Underfitting): The model makes strong incorrect assumptions about the data. ● High Variance (Overfitting): The model is overly sensitive to small fluctuations in the training set. ● The Tradeoff: As we increase model complexity, Bias decreases but Variance increases. The goal is to find the "Sweet Spot" where the total error is minimized. **3. Methods to Prevent Overfitting (2 Marks)** 1. Regularization (L1/L2): Adds a penalty term to the cost function that discourages the model from assigning too much importance to any single feature (shrinks coefficients). 2. Cross-Validation: Using techniques like K-fold cross-validation helps ensure the model generalizes well across different subsets of data rather than memorizing one specific set. **Confusion Matrix and Evaluation Metrics** **1. Definition (1 Mark)** A **Confusion Matrix** is a N * N table used to evaluate the performance of a classification model, where N is the number of target classes. It compares the **Actual** target values with those **Predicted** by the model. **2. Deriving Formulas (3 Marks)** Given the components: **TP** (True Positive), **TN** (True Negative), FP (False Positive), and FN (False Negative). ● Accuracy: The overall correctness of the model. ● Precision: The accuracy of positive predictions. ● Recall (Sensitivity): The ability of the model to find all actual positive cases. **3. Importance of Recall in Disease Detection (2 Marks)** In "Disease Detection" (e.g., Cancer or COVID-19), the cost of a **False Negative** (telling a sick person they are healthy) is much higher than a **False Positive** (telling a healthy person they might be sick). ● Accuracy is misleading ● Recall is critical	CO4	4B	K4

Part- C

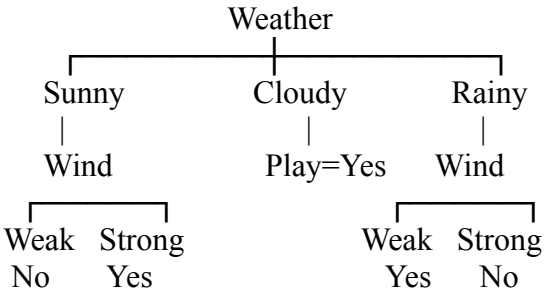
<table>
<tr><th>Q.No</th><th>Questions</th><th>CO’s</th><th>Unit</th><th>Bloom’s Level</th></tr>
<tr><td>1.</td><td>Construct the decision tree for the below dataset
<table>
<tr><th>Day</th><th>Weather</th><th>Temperature</th><th>Humidity</th><th>Play?</th></tr>
<tr><td>1</td><td>Hot</td><td>High</td><td>Weak</td><td>No</td></tr>
<tr><td>2</td><td>hot</td><td>High</td><td>Weak</td><td>Yes</td></tr>
<tr><td>3</td><td>Mild</td><td>Normal</td><td>Strong</td><td>Yes</td></tr>
<tr><td>4</td><td>Mild</td><td>High</td><td>Strong</td><td>Yes</td></tr>
<tr><td>5</td><td>Mild</td><td>High</td><td>Strong</td><td>No</td></tr>
<tr><td>6</td><td>Cold</td><td>Normal</td><td>Strong</td><td>No</td></tr>
<tr><td>7</td><td>Mild</td><td>High</td><td>Weak</td><td>Yes</td></tr>
<tr><td>8</td><td>Hot</td><td>High</td><td>Strong</td><td>No</td></tr>
<tr><td>9</td><td>Hot</td><td>Normal</td><td>Weak</td><td>Yes</td></tr>
<tr><td>10</td><td>Mild</td><td>High</td><td>Strong</td><td>No</td></tr>
</table></td><td>CO4</td><td>4C</td><td>K4</td></tr>
<tr><td>1 Ans</td><td>To construct a decision tree for the provided dataset, we will use the ID3 algorithm, which relies on Information Gain (using Entropy) to determine the best splitting attributes.
Step 1: Dataset summary (2 marks)
Total instances = 10
● Yes = 5
● No = 5
So initial entropy:

Step 2: Choose the root attribute (highest Information Gain) (3 marks)
After calculating Information Gain for all attributes (Weather, Temperature, Humidity, Wind), the attribute with highest IG is:
1. Weather
So, Weather becomes the root node.

Step 3: Split based on Weather (5 marks)
Weather = Sunny
Records: Day 1, 3, 8
● Yes = 1
● No = 2
Next best attribute here → Wind
● Wind = Weak → No
● Wind = Strong → Yes

2. Weather = Cloudy
Records: Day 2, 4, 9
● Yes = 3
● No = 0
All Yes (leaf node)

3. Weather = Rainy</td><td>CO4</td><td>4C</td><td>K4</td></tr>
</table>

	Records: Day 5, 6, 7, 10 ● Yes = 1 ● No = 3 Next best attribute → **Wind** ● Wind = Weak → **Yes** ● Wind = Strong → **No** ● **Final Decision Tree(5 Marks)**  			
2.	The values of x and their corresponding values of y are shown in the table below. x: 1 2 3 4 5 6 7 y: 3 4 5 5 6 8 10 1)Analyze the relationship between x and y and derive the least squares regression line y=ax+b. (12) 2)Estimate the value of y when x = 10 (3)	CO4	4C	K4
	1.Analysis and Derivation of the Regression Line (11 Marks) Step 1: Calculate the slope (a)=1.07 Step 2: Calculate the intercept (b)=1.57 The Least Squares Regression Line is: Y=1.07X + 1.57 **2.Estimate the value of y when x = 10 (4 Marks)** the predicted value of Y=12.29	CO4	4C	K4

Unit- 5

Part A

Q.No	Questions	CO's	Unit	Bloom's Level
1.	Define data visualization and state its importance in data science.	CO5	5A	K1
1 Ans	● **Definition:** Data visualization is the graphical representation of information and data using visual elements like charts, graphs, and maps. ● **Importance:** It is crucial in data science because it helps stakeholders understand trends, outliers, and patterns in data[2]. It translates complex datasets into an accessible format, facilitating better decision-making and storytelling.	CO5	5A	K1
2.	List any two principles of effective data visualization.	CO5	5A	K1

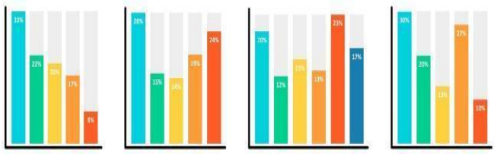
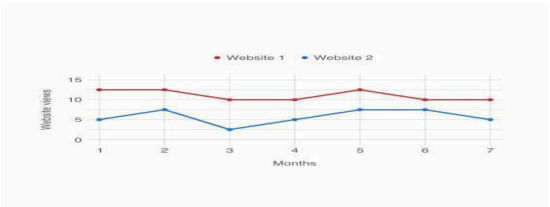
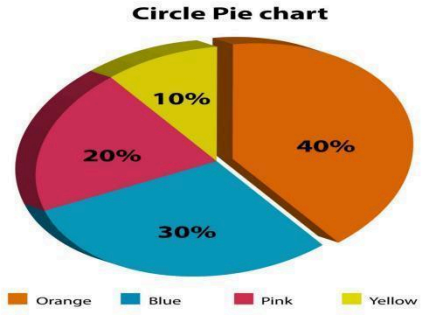
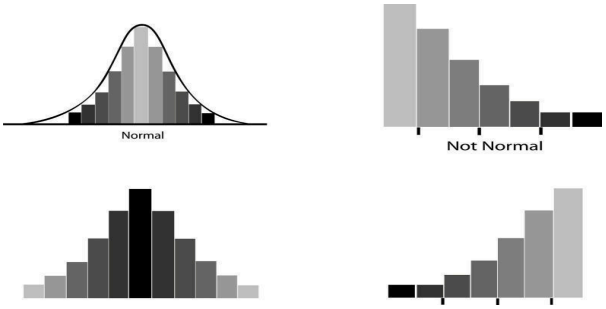
2 Ans	● **Clarity:** The visualization should clearly communicate the data without ambiguity or unnecessary clutter (chartjunk). ● **Accuracy:** The visual representation must accurately reflect the underlying data without distorting the scale or relationships.	CO5	5A	K1
3.	Differentiate between qualitative and quantitative data with an example.	CO5	5A	K3
3 Ans	● **Qualitative Data:** Describes categories or qualities that cannot be counted or measured numerically. ● *Example:* Customer feedback sentiment (Positive, Neutral, Negative) or car colors (Red, Blue, Black). ● **Quantitative Data:** Represents measurable quantities or numbers. ● *Example:* The price of a house ($300,000) or the temperature of a city (25°C).	CO5	5A	K3
4.	Identify two situations where a bar chart is more suitable than a line chart.	CO5	5A	K2
4 Ans	● **Comparing Categories:** When comparing discrete groups or categories (e.g., sales by department: Marketing, Sales, HR) rather than continuous data. ● **Non-Time Series Data:** When the data does not have a sequential or time-based relationship (e.g., comparing the population of different countries). ● (Note: The uploaded Python book discusses creating Bar Plots for categorical data visualization.)	CO5	5A	K2
5.	Interpret what a right-skewed histogram indicates about a dataset.	CO5	5A	K2
5 Ans	● A right-skewed (or positively skewed) histogram indicates that the tail of the distribution extends to the right. ● **Interpretation:** This means that the majority of the data points are concentrated on the left side (lower values), with a few high values pulling the mean to the right. typically, in a right-skewed distribution, the **Mean > Median**. ● *Example:* Income distribution is often right-skewed, where most people earn a moderate amount, but a few high earners create a long tail to the right.	CO5	5A	K2
6.	Identify two advantages of using box plots for data interpretation.	CO5	5A	K2
6 Ans	● Efficient Outlier Detection: Box plots explicitly identify outliers by plotting them as individual points beyond the "whiskers." This makes it much easier to spot anomalies that might be hidden in a standard histogram or bar chart. ● Visualizing Dispersion and Skewness: They provide a clear view of the Interquartile Range (IQR) and the Median. By looking at the	CO5	5A	K2

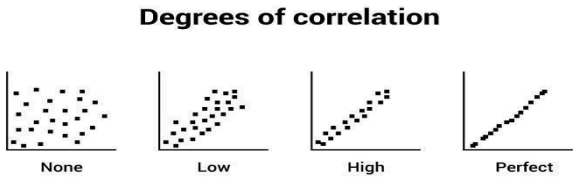
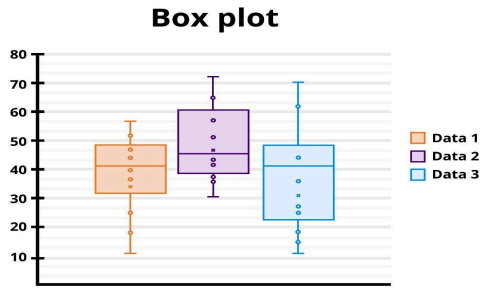
	position of the median line within the box and the length of the whiskers, an analyst can immediately tell if the data is symmetrical or skewed without performing complex calculations.			
7.	Define data ethics in the context of data science.	CO5	5A	K1
7 Ans	Definition: Data ethics refers to the moral obligations and responsible practices regarding the collection, analysis, and dissemination of data. It ensures that data science processes respect privacy, fairness, transparency, and do not cause harm to individuals or society.	CO5	5A	K1
8.	Give two examples of unethical use of data.	CO5	5A	K2
8 Ans	Data Bias/Discrimination: Using biased historical hiring data to train an AI model that systematically rejects female candidates or candidates from specific demographics. Unauthorized Data Sharing: A company selling sensitive user data (like health records or location history) to third-party advertisers without the users' explicit consent or knowledge	CO5	5A	K2
9.	Write the concept of data privacy with one real-world example.	CO5	5A	K1
9 Ans	Concept: Data privacy concerns the proper handling, processing, and storage of personal information, ensuring that individuals have control over their own data and that it is protected from unauthorized access. Example**:** A hospital implementing strict access controls and encryption for patient records to comply with regulations like HIPAA or GDPR, ensuring that a patient's medical history is only visible to their treating doctors.	CO5	5A	K1
10.	Give two common data security threats in data science projects.	CO5	5A	K1
10 Ans	Data Breaches: Unauthorized access where sensitive data is stolen or copied by a hacker or malicious actor. Insider Threats: Risks posed by employees or authorized users who misuse their access privileges to leak, alter, or steal data, either maliciously or accidentally.	CO5	5A	K1
11.	Apply the principle of “simplicity” in designing a dashboard.	CO5	5A	K3
11 Ans	The principle of simplicity involves removing non-essential elements (chart junk) to focus on key insights. This is achieved by using clean layouts, limiting the color palette to highlight important data, and ensuring that the most critical KPIs are visible without requiring the user to scroll or filter extensively.	CO5	5A	K3
12.	Apply an appropriate chart type to visualize sales trends over time. Justify your choice.	CO5	5A	K3

12 Ans	A Line Chart is the most appropriate choice. Justification: Line charts are designed to show continuity. By connecting data points over chronological intervals (days, months, or years), they allow the viewer to quickly identify patterns, seasonal spikes, or downward trends that would be harder to see in a table or bar chart.	CO5	5A	K3
13.	Interpret a scatter plot showing a negative correlation between two variables.	CO5	5A	K3
13 Ans	In a scatter plot, a negative correlation is represented by a data trend that slopes downward from left to right. This indicates an inverse relationship: as the value of the independent variable (x) increases, the value of the dependent variable (y) decreases.	CO5	5A	K3
14.	Explain the need for secure data storage in data science applications.	CO5	5A	K2
14 Ans	Secure data storage is vital to protect data integrity and privacy. In data science, unauthorized access or data breaches can lead to the exposure of sensitive personal information (PII), leading to legal consequences (like GDPR violations) and the loss of organizational trust or competitive advantage.	CO5	5A	K2
15.	Identify two real-world applications of data visualization.	CO5	5A	K2
15 Ans	● Healthcare: Using heatmaps to track the geographic spread of a virus or disease outbreak in real-time. ● Finance: Using candlestick charts or line graphs to analyze stock market fluctuations and assist in investment decision-making	CO5	5A	K2

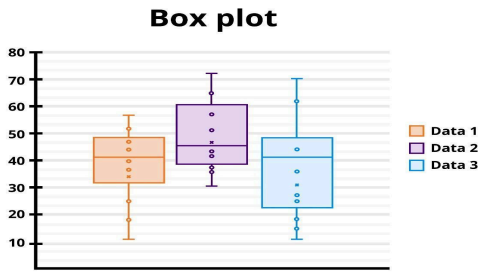
Part- B

Q.No	Questions	CO's	Unit	Bloom's Level
1.	Analyze the principles of effective data visualization and apply them to design a sales performance chart.	CO5	5B	K4
1 Ans	**1. Principles of Effective Data Visualization (6) marks)** Effective data visualization transforms raw data into meaningful insights. To achieve this, several key principles (often derived from experts like Edward Tufte) must be followed: ● Clarity and Simplicity (The "Data-Ink Ratio") ● Accuracy and Integrity ● Context and Relevance ● Consistency **2. Application: Designing a Sales Performance Chart (7 marks)** Scenario: We need to design a chart to show a company's sales performance over the last 12 months compared to the previous year.	CO5	5B	K4

	Applied Strategy: Step 1: Selection of the Right Chart Type Step 2: Simplifying the Design (Reducing Clutter) Step 3: Ensuring Accuracy Step 4: Using Colour for Focus Step 5: Adding Context			
2.	Analyse different types of charts used for data representation and justify their suitability for various data types.	CO5	5B	K4
2 Ans	**1. Bar Chart (Bar Graph)(2 marks)** ● **Analysis:** Uses rectangular bars where the length or height of the bar is proportional to the value it represents. It can be vertical (column chart) or horizontal. ● **Suitable Data Types: Categorical (Nominal/Ordinal) vs. Quantitative.** ● **Justification:** ● **Comparison:** ● **Flexibility:**  **2. Line Chart (2 marks)**  **3. Pie Chart (and Donut Chart) (2 marks)**  **4. Histogram (3 marks)**  ● **5. Scatter Plot (2 marks)**	CO5	5B	K4

	 **6. Box Plot (Box-and-Whisker Plot) (2 marks)** 			
3.	Evaluate the ethical issues in data science with suitable real-world examples.	CO5	5B	K5
3 Ans	Data science offers immense power to transform industries, but it also raises significant ethical challenges. These issues generally fall into several key categories: privacy, bias, transparency, and accountability. Here are the primary categories of ethical issues with real-world examples: **1. Privacy and Surveillance (3 marks)** ● The Issue ● Real-World Example **2. Bias and Discrimination (Algorithmic Fairness)(2 marks)** ● The Issue ● Real-World Example: Amazon's AI Recruiting Tool. **3. Transparency and Explainability (The "Black Box" Problem)(2 marks)** ● The Issue ● Real-World Example: **4. Informed Consent and Data Ownership (3 marks)** ● The Issue ● Real-World Example **5. Accountability and Responsibility(3 marks)** ● The Issue ● Real-World Example: Uber Self-Driving Car Crash (2018).	CO5	5B	K5
4.	Analyse the challenges related to data privacy and security in modern data-driven organizations.	CO5	5B	K4
4 Ans	**1. Data Privacy Challenges (4 marks)** Privacy challenges revolve around the ethical and legal handling of personal information.	CO5	5B	K4

	● Consent and Transparency ● De-identification and Re-identification ● Compliance with Global Regulations **2. Data Security Challenges (4 marks)** Security challenges focus on the technical defense mechanisms against malicious attacks and internal errors. ● Data Breaches and Cyberattacks ● Insider Threats ● Insecure Interfaces and APIs **3. The "Big Data" Paradox (5 marks)** ● The Challenge: The very nature of Big Data (Volume, Velocity, Variety) makes security harder. ● Analysis: Traditional security tools cannot scan petabytes of data in real-time. Encryption also slows down processing speed. Organizations are often forced to choose between Performance (fast analysis) and Security (encrypted, locked-down data), often leading to dangerous compromises.			
5.	Prepare the process of interpreting charts and graphs for business decision-making.	CO5	5B	K5
5 Ans	**1. Phase I: Structural Analysis (The "What") (4 marks)** ● Check the Metadata ● Examine the Axes and Scales **2. Phase II: Pattern Recognition (The "How") (3 marks)** ● Trend Identification ● Spotting Outliers (Anomalies) ● Seasonality **3. Phase III: Contextual Synthesis (The "Why") (3 marks)** Data does not exist in a vacuum. This step connects the visual pattern to external business factors. ● Correlation vs. Causation ● Benchmarking **4. Phase IV: Decision Formulation (The "Now What") (3 marks)** The final and most important step is converting the insight into a business decision. ● Strategic Decisions (Long-term) ● Tactical Decisions (Short-term) ● Risk Mitigation	CO5	5B	K5
6.	Analyze the effectiveness of different graphical techniques in identifying outliers and patterns in data.	CO5	5B	K4
	Exploratory Data Analysis (EDA) relies heavily on graphical techniques to visually detect anomalies (outliers) and underlying structures (patterns) that summary statistics (like mean or standard deviation) might miss.[1] Different charts differ in their sensitivity and effectiveness for these tasks.			

6 Ans	Here is an analysis of the most effective graphical techniques: **1. Box Plots (Box-and-Whisker Plots) (4marks)** The Box Plot is arguably the most effective standardized tool specifically designed for outlier detection.  **2. Scatter Plots (3 marks)** Scatter plots are the gold standard for analyzing the relationship between two numerical variables (bivariate data). **3. Histograms (3 marks)** Histograms provide a view of the frequency distribution of a single continuous variable. ● might contain two different groups (e.g., heights of men and women mixed together). **4. Line Charts (3 marks)** Line charts are specialized for Time-Series data.	CO4	5B	K4
7.	Outline a real-world case study where data visualization played a key role in insights generation.	CO5	5B	K4
7 Ans	**1. The Problem: A Deadly Mystery (4 marks)** In September 1854, a severe outbreak of cholera hit the Soho district of London.[2] In just three days, 127 people died.[3] At the time, the prevailing scientific theory was the "Miasma Theory," which suggested that diseases like cholera were spread by "bad air" or foul smells from decaying organic matter. Dr. John Snow, a local physician, was skeptical of this theory.[5] He suspected cholera was waterborne, but he lacked the "smoking gun" to prove it to local officials. **2. The Data Visualization Approach (3 marks)** Snow decided to map the deaths to see if a spatial pattern emerged.[6] He used a simple but revolutionary technique: ● The Map: He took a street map of the Soho neighborhood. ● The Data Points: For every death recorded, he drew a small black bar at the specific street address where the victim lived. ● The Variables: He also marked the locations of the eleven municipal water pumps in the area. **3. Insights Generated Through Visualization (3 marks)** When the data was plotted visually, the insight was immediate and undeniable:	CO5	5B	K4

	● Clustering: The black bars (deaths) were heavily concentrated around the intersection of Broad Street and Cambridge Street. ● The Correlation: The visual epicenter of the deaths was the Broad Street water pump.[7] **4. The Impact and Outcome(3 marks)** ● The visualization provided the narrative force needed to convince the Board of Guardians. Despite their skepticism of his germ theory, the visual evidence was so compelling that they agreed to remove the handle of the Broad Street pump.			
8.	Evaluate emerging trends in data science and their influence on industry and society.	CO5	5B	K5
8 Ans	**1. The Transition to Agentic & Multimodal AI[2] (2 marks)** The primary shift in 2026 is moving away from "chatbots" toward Agentic AI.[3] These systems do not just answer questions; they execute multi-step goals autonomously across different software environments.[4] Influence on Industry: In logistics, an AI agent can identify a supply chain disruption, calculate the cost of alternative routes, and book new shipping containers without human intervention. Influence on Society: Personal AI assistants have become "life managers," capable of handling complex administrative tasks like coordinating healthcare appointments, managing household energy consumption, and filing taxes. **2. Edge Intelligence and Real-Time Analytics (3 marks)** ● Impact Comparison **3. The Democratization of Data (Natural Language Querying) (3 marks)** Data science is no longer a "walled garden" for those who code in Python or R. The rise of Augmented Analytics allows non-technical employees to perform complex data mining using natural language.[7] ● Influence: ● Societal Shift **4. Ethical AI and "Sovereign" Data Regulation(3 marks)** ● Algorithmic Transparency: Companies must now be able to explain *why* an AI denied a loan or rejected a job application. **5. Sustainability: "Green AI"(2 marks)** The massive energy consumption of 2024–2025's AI models has led to a 2026 trend of Resource-Efficient Machine Learning. ● Smaller Models: Instead of 1-trillion-parameter "God models," industries are shifting to Domain-Specific Models (e.g., a small, highly	CO5	5B	K5

	efficient model trained only on legal case law) which require 90% less energy to run.[11]			

Part- C

Q.No	Questions	CO's	Unit	Bloom's Level
1.	Design a complete data visualization strategy for a healthcare dataset, ensuring ethical use, data privacy, and clarity of interpretation.	CO5	5C	K6
1.Ans	**1. Governance & Ethical Framework (3 marks)** Before a single pixel is rendered, you must establish the ethical "guardrails" of the project. ● Privacy by Design ● The "So What?" Principle. ● Bias Mitigation **2. Technical Data Architecture (3 marks)** Visualizations are only as good as the underlying pipeline. ● De-identification Layer: ● Role-Based Access Control (RBAC ● Data Provenance: **3. Visualization Design Strategy (3 marks)** Healthcare data is often longitudinal (time-series) and multi-dimensional. Clarity is the primary goal. Choosing the Right Visuals ● Survival & Outcomes ● Population Health ● Patient The "Clinical Dashboard" Hierarchy **4. Clarity of Interpretation & Literacy (3 marks)** ● Standardized Color Palettes: ● Uncertainty Visualization: ● Plain Language Annotations: **5. Continuous Evaluation (3 marks)** A visualization strategy is not static. It requires an Audit Loop: ● User Feedback ● Drift Monitoring	CO5	5C	K6
2.	Design an end-to-end data science workflow for a smart city application, highlighting visualization, privacy, and security considerations.	CO5	5C	K6
2 Ans	**1. The End-to-End Workflow Architecture (5marks)** Stage 1: Data Acquisition & Edge Computing Stage 2: Ingestion & Privacy Scrubbing Stage 3: Feature Engineering & Federated Learning Stage 4: Modeling & Model Registry **2. Visualization & Interface Strategy (5marks)** Visualization in a smart city serves two masters: Operators (who need precision) and Citizens (who need transparency). ● For City Operators (Control Centers) ● For the Public (Citizen Dashboards)	CO5	5C	K6

	3. Ethical Considerations (The 2026 Standard) (5marks) ● Democratization: Allow citizens to "Opt-out" of certain non-essential data collection via a central city app. ● Bias Audits: Annual audits to ensure AI resource allocation (like trash pickup or police patrols) isn't unfairly targeting specific socio-economic neighborhoods. ● The "Human-in-the-loop": Ensure that high-impact decisions (e.g., shutting down a bridge) always require a human override from a city engineer.			